"ARRAYS" IN JAVASCRIPT (VERY SHORT INTRODUCTION)

Generally speaking, an array is a fixed-size group of variables (called ***elements***) in which an individual element is identified by its position in the array, relative to the first element.

lotteryNumbers

23	**12**	**7**	**9**	**36**	**2**	**16**

To refer to a particular element in the array, we specify the name of the reference and the ***index*** (= position number) of the element in the array. An index must be a non-negative integer (and can be any expression that evaluates to a positive integer). Square brackets are used in array-indexing expressions.

lotteryNumbers

23	**12**	**7**	**9**	**36**	**2**	**16**
0	1	2	3	4	5	6

In the example below, an array of seven numbers is created.

```
var lotteryNumbers = [23, 12, 7, 9, 36, 2, 16];

console.log (lotteryNumbers[4]);  // Prints 36

console.log (lotteryNumbers[0]);  // Prints 23
```

NB! Indexing of array elements ***starts from zero*** in many popular programming languages.

Typically, a ***for loop*** is used to iterate through array elements. The for loop below iterates through the array and prints the value of each array element.

```
for (var i = 0; i <= 6; i++) {
    console.log (lotteryNumbers[i]);
}
```

An array has also the property **length** that gets the number of elements in the array.

```
console.log (lotteryNumbers.length);     // Prints 7
```

The number of elements in an array may be zero, in which case the array is said to be empty.

NB! The last element of an array is at the index equal to the value of the array's length property minus one.

```
for (var i = 0; i < lotteryNumbers.length; i++) {
    console.log (lotteryNumbers[i]);
}
```

In the example above, the ***length*** property is used in the loop condition.

In JavaScript, "arrays" are list-like collections. JavaScript "arrays" are ***dynamic data structures***. 'Dynamic' means that you do not have to specify the size of the data structure when you create it, and the structure grows and shrinks dynamically when you add new data in it or remove data from it.

In JavaScript, the easiest way of creating an array is to use an ***array initializer*** (array literal). An array initializer is a comma-separated list of values in square brackets.

Then you can add new elements to the end of the array by calling the ***push*** method as in the example below.

```
var attendees = [];                    // Creates an empty array

console.log(attendees.length);         // Prints 0

attendees.push("Frank");       // Adds a new element to the end of the array

attendees.push("Susan");       // Adds a new element to the end of the array

attendees.push("John");        // Adds a new element to the end of the array

attendees[1] = "Alice";        // Changes the value of the array element


for (var i = 0; i < attendees.length; i++) {
    console.log (attendees[i]);
}
```

The for loop above prints the following:

```
Frank
Alice
John
```

There are many useful built-in operations for arrays. For example, you can sort an array as in the example below.

```
attendees.sort();

for (var i = 0; i < attendees.length; i++) {
    console.log (attendees[i]);
}
```

The for loop above prints the following:

```
Alice
Frank
John
```

To see more information
- http://www.w3schools.com/js/js_arrays.asp
- https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Array